Mariana Canelas Pais

Medical doctor · Health informatics and data engineering · linkedin.com/in/mariana-canelas-pais-304489175 · ORCID 0000-0002-6487-313X

01 Education and certifications

PhD candidate in Health Data Science
Faculty of Medicine, University of Porto · since 09/2023

MSc in Medical Informatics — 19.0 / 20
Faculty of Medicine, University of Porto · 09/2021 – 12/2023
Dissertation: *Visual Viper: a portable visualization library for streamlined scientific communications*

Board certification in General Surgery — 19.50 / 20
Centro Hospitalar Universitário de São João / Portuguese Medical Association · 07/2020

MD — Integrated Master's Degree in Medicine — 15.19 / 20
Faculty of Medicine, University of Coimbra · completed 06/2012

Certificate of Proficiency in English, level C2 — Grade B
University of Cambridge · 2006

PROFESSIONAL CERTIFICATION

Certificate of Pedagogical Competences (CCP)
ENA — Escola de Negócios e Administração · issued 10/2021

openEHR — professional training course, 25 hours · final grade 17 / 20
Ciência e Letras — Formação Especializada · completed 11/12/2025

02 Professional experience

MTG Research and Development Lab — Porto, Portugal

Principal Data Engineer · since 01/2026

> Leads the team's data engineering: ETL pipelines and the harmonisation of clinical data to common data models such as the OMOP CDM.

Data Science Director · since 03/2024

> - Design and delivery of scientific studies in healthcare, including Real World Evidence studies.
> - Management of scientific projects: teams, deliverables and documentation.
> - Statistical analysis, interpretation and communication of results.
> - Harmonisation of electronic health record data to common data models such as the OMOP CDM.
> - Engagement with clinical teams, key opinion leaders and client stakeholders in study design.
> - Python, R, SQL and AWS services.

Senior Data Scientist · 01/2023 – 02/2024

Faculty of Medicine, University of Porto

Invited Assistant · since 2022/2023 · fixed-term contract, part-time

> Teaching in the Integrated Master's Degree in Medicine, across the three DECIDES course units: *Decision, Data and Statistics in Health* and *Decision, Data and Evidence in Health* (2nd year), and *Decision, Data and Digital Health* (4th year).

Voluntary Assistant · 09/2017 – 12/2019

> Teaching in Endocrine Diseases, Nutrition and Metabolism, 4th year of the Integrated Master's Degree in Medicine.

Research Fellow, full-time · 01/2022 – 12/2022

> **IMAGE** project — *Individualized gastric adenocarcinoma early diagnosis and improved patients' survival: from liquid biopsies to a comprehensive management approach* (NORTE-01-0145-FEDER-000050). Definition of research variables and data dictionaries; REDCap data collection solution, including form design and user training; preparation of the project's Data Management Plan; coordination with the clinical and laboratory research teams.

Centro Hospitalar Universitário de São João — Porto, Portugal

General Surgery Resident · 01/2014 – 08/2020

> Specialty training in General Surgery, completed with **19.50 / 20** in the final board examination (02/07/2020). Elective and emergency surgical activity, with optional rotations in Intensive Care Medicine, Paediatric Surgery, Vascular Surgery, and Plastic and Reconstructive Surgery.
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> - **Research:** around 30 congress communications as first author and co-author; participation in the international multicentre RIFT Study Group (2017), published in the *British Journal of Surgery*; update of the Breast Surgery Protocol of the CHUSJ Breast Centre (2016).
> - **Teaching:** undergraduate teaching at FMUP, including the Experimental Surgery classes; speaker at the CHUSJ *Surgeon to Surgeon* minimally invasive surgery courses (2014, 2015) and the Colorectal *Minifellow* (2017).
> - **Organisation:** organising committee of the IV, V and VI International Congresses of Surgical Residents, Portuguese Medical Association — Northern Branch (2014–2016); resident representative for the department (2019).

Centro Hospitalar de Vila Nova de Gaia/Espinho, E.P.E.

Medical Intern · 01/2013 – 12/2013

> Rotations in Internal Medicine, General Surgery, Otorhinolaryngology, Cardiology, Paediatrics and General Practice.

03 Training activity

Certified trainer, holder of the Portuguese Certificate of Pedagogical Competences. Delivers professional training on artificial intelligence applied to healthcare — including generative AI and conversational systems — to clinical audiences and to professionals from other sectors.

04 Thesis supervision

Co-supervisor, MSc in Health Informatics, Faculty of Medicine, University of Porto. Official record: `sigarra.up.pt/fmup/pt/TESES.TESES_ORIENTADOR?p_docente=671186`

From Real-World Data to Clinical Visualization: Developing a Data Model for Wound Care — Hélder Avelino de Sousa Palheira, defended 09/12/2025. Clinical modelling in openEHR (archetypes and templates), validation on EHRbase, and dashboards for the secondary use of chronic wound care data.

Metadata in Medical Imaging: unlocking its potential for evidence generation in clinical research — Susana Cristina Lemos Sebastião, defended 17/12/2025. Secondary use of DICOM metadata and its mapping to the OMOP CDM, using the OHDSI tooling (White Rabbit, Rabbit in a Hat).

05 Professional bodies

Vice-President · E-Mais — Association for Health Information Systems · since 03/2025
Portuguese association for health information systems, and the Portuguese representative to the *European Federation for Medical Informatics* (EFMI).

Member · Portuguese Society of Surgery

06 Scientific output

6.1 Peer-reviewed journal articles

1. Fonseca Sousa J, Rocha B, Silva M, **Canelas-Pais M**, Coelho A, Guimarães I, Queirós V, Baptista R. CardioGAP+: bridging the gap between patient identification and therapy implementation in cardiovascular risk. *Frontiers in Cardiovascular Medicine*. 2026;13:1881174.
2. Pineda-Moncusí M, Rekkas A, [...] **Canelas-Pais M**, [...] Prieto-Alhambra D, Rijnbeek PR, Burkard T (Drug Shortages EHDEN Study Group). Changes in use and utilisation patterns of drugs with reported shortages between 2010 and 2024 in Europe and North America: a network cohort study. *The Lancet Public Health*. 2025;10(10):e835–e847.
3. Dias Costa P, Bessa JP, **Canelas Pais M**, Ferreira-Santos D, Montenegro Sá F, Monteiro-Soares M, Hipólito-Reis A, Martins Oliveira M, Pereira Rodrigues P. Prediction of response to cardiac resynchronization therapy using electrocardiographic criteria: a systematic review. *Revista Portuguesa de Cardiologia*. 2025;44(9):561–576.
4. Gavina C, Carvalho DS, Afonso-Silva M, Costa I, Freitas AS, **Canelas-Pais M**, Lourenço-Silva N, Taveira-Gomes T, Araújo F. Lipid-lowering prescription patterns after a non-fatal acute coronary syndrome: a retrospective cohort study. *International Journal of Cardiology: Cardiovascular Risk and Prevention*. 2025;25:200385.
5. PTSurg — Portuguese Surgical Research Collaborative. Ambulatory inguinal hernia repair in Portugal: a multicenter prospective cohort study. *Updates in Surgery*. 2025;77(8):2591–2600.
6. Gavina C, Borges A, Afonso-Silva M, Fortuna I, **Canelas-Pais M**, Amaral R, Costa I, Seabra D, Araújo F, Taveira-Gomes T. Patients' health care resources utilization and costs estimation across cardiovascular risk categories: insights from the LATINO study. *Health Economics Review*. 2024;14(1):73.
7. Gavina C, Seabra Carvalho D, Afonso-Silva M, Brandão Abreu D, **Canelas-Pais M**, Taveira-Gomes T, Araújo F. Long-term cardiovascular outcomes in a population-based multicentric cohort of northern Portugal: validation of the ESC/EAS prognostic risk classification. *Journal of Clinical Lipidology*. 2024;18(4):e562–e571.
8. Gavina C, Seabra-Carvalho D, Aguiar C, Anastassopoulou A, Teixeira C, Ruivo JA, Almeida É, Luz-Duarte L, Corte-Real A, **Canelas-Pais M**, Taveira-Gomes T. Characterization and LDL-C management in a cohort of high and very high cardiovascular risk patients: the PORTRAIT-DYS study. *Clinical Cardiology*. 2024;47(1):e24183.
9. Ferreira JP, Taveira-Gomes T, **Canelas-Pais M**, Phan P, Bernardo F, Andersson Sundell K, Almeida M, Martinho H, Gavina C. Missed opportunities in the diagnosis of heart failure: a real-world assessment. *ESC Heart Failure*. 2023;10(6):3438–3445.
10. Gavina C, Araújo F, Teixeira C, Ruivo JA, Corte-Real AL, Luz-Duarte L, **Canelas-Pais M**, Taveira-Gomes T. Sex differences in LDL-C control in a primary care population: the PORTRAIT-DYS study. *Atherosclerosis*. 2023;384:117148.
11. Rei A, Ortigão R, **Pais M**, Afonso LP, Pimentel-Nunes P, Dinis-Ribeiro M, Libânio D. Metachronous lesions after gastric endoscopic submucosal dissection: first assessment of the FAMISH prediction score. *Endoscopy*. 2023;55(10):909–917.
12. Rafael F, Rodrigues MD, Bellver J, **Canelas-Pais M**, Garrido N, Garcia-Velasco JA, Soares SR, Santos-Ribeiro S. The combined effect of BMI and age on ART outcomes. *Human Reproduction*. 2023;38(5):886–894.

13. Simões JFF, Azevedo J, da Costa JP, Cunha M, Ferreira JJ, Guimarães A, [...] (**Pais M**). Predictors of low quality of life after open inguinal hernia repair using the EuraHS-QoL score: prospective multicentric cohort study across 33 hospitals. *Hernia*. 2022;26(1):225–232.
14. Evaluation of appendicitis risk prediction models in adults with suspected appendicitis. *British Journal of Surgery*. 2020;107(1):73–86. *(RIFT Study Group — international multicentre study)*
15. **Pais MC**, Maia T, Peixoto A, Meira L, Albuquerque A, Lopes J, Capela J. Diarrhea as a form of presentation of medullary thyroid carcinoma. *Porto Biomedical Journal*. 2018;3(3):e18.
16. Sousa F, Duarte A, Fernandes C, Morais M, **Pais M**, Correia da Silva P, Costa-Maia J. TAMIS: our initial experience. *Revista Portuguesa de Coloproctologia*. 2018;15(1):23–28.

6.2 Book chapter

17. **Canelas-Pais M**, Coutinho-Almeida J, Araujo SM, Rafael FM, Cruz-Correia R, Rodrigues PP. Artificial intelligence in healthcare. In: *Artificial Intelligence for Drug Product Lifecycle Applications*. Elsevier, 2025:235–271.

6.3 Preprint

18. Pineda-Moncusí M, Rekkas A, [...] **Canelas-Pais M**, [...]. Trends of use of drugs with suggested shortages and their alternatives across 52 real world data sources and 18 countries in Europe and North America. *medRxiv* 2024.08.28.24312695.

6.4 Selected conference communications

19. **Canelas-Pais M**, da Silva RF, Duarte I, Rodrigues PP, Fonseca JA, Cruz-Correia R. Integration of artificial intelligence in medical education: developing a framework for curriculum enhancement. *MEDINFOR VI*. 2023.
20. Silva R, Gomes S, **Canelas-Pais M**, Sá-Sousa A, Taveira-Gomes T. Lessons learned from implementing OMOP ETLs across EHR and registry-based data sources. OHDSI Europe 2026.
21. Silva R, Lopes R, Luz R, Sá-Sousa A, Jácome C, **Canelas-Pais M**. Bridging the gaps in clinical timelines: unlocking OMOP's relationship potential with FACT_RELATIONSHIP. OHDSI Europe 2025.
22. Seabra Carvalho D, Gavina C, Afonso-Silva M, **Canelas-Pais M**, Araújo F. Low-density lipoprotein cholesterol reduction and short-term incidence of ASCVD in the population-based cohort study LATINO. *European Heart Journal*. 2023;44(Suppl. 2):ehad655.2510.
23. Gavina C, Teixeira C, Ruivo J, Anastassopoulou A, **Canelas-Pais M**, et al. Gender differences in attainment of LDL-C goals independently of treatment intensity. *Atherosclerosis*. 2023;379:S124–S125.

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07 Skills

TECHNICAL SKILLS

R · Python · SQL · Git · LaTeX · HTML · CSS

Amazon Web Services (S3, Lambda)

Common data models (OMOP CDM) · openEHR · ETL over electronic health records · REDCap · Real World Evidence · data management plans · data dictionaries

LANGUAGES

Portuguese — native

English — C1–C2, Certificate of Proficiency in English

Spanish — A1–A2